

Question	Marking Guidance	Mark	Comments
02.1	Correlation coefficient and because looking for correlation / relationship / association between two variables / between B cells destroyed and CD20;	1	Accept Pearson and Spearman Accept factor for variable Wrong test or wrong reason = 0 marks
02.2	1. The more CD20 (on B cells), higher the percentage of / more B cells destroyed / more effective it is; 2. (At best) only destroys (about) 80% of B cells OR In no cases are all B cells killed; 3. Don't know % / proportion of cancer cells killed; 4. Won't cure CLL / cancer / slows but doesn't stop CLL / cancer; 5. Little effect below (about) 5 CD20 on cells;	3 max	1. Ignore ref. to 'positive correlation' unqualified Ignore ref. to correlation vs. causation Ignore ref. to effects on the immune system 5. Ignore ref. to little effect where little CD20
02.3	1.8×10^8 ;; If correct difference but expressed in non-standard form, award 1 mark;	2	Award 1 mark if answer given as 1.8×10^{-8}

02.4	1. Mutation changes the tertiary structure / amino acid sequence of transcription factor; 2. Transcription factor not complementary to / cannot bind to CD20 gene / CD20 DNA; 3. Little / less / not enough / no <u>mRNA</u> for CD20 produced OR Little / less / not enough / no (mRNA for) CD20 translated / produced; 4. (Not enough CD20 so) nothing / little for Rituximab to bind to, so few / no B cells destroyed;	3 max	1. and 2. Do not accept unqualified statements about non-functional transcription factor – this is in stem of question 1. and 2. Answers must be in context of transcription factor, not CD20, or generic statements 2. Accept TF cannot bind to promoter (on DNA) 4. Accept converse for cells with a lot of CD20 4. Accept lower median percentage for fewer cells destroyed
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